

GestureShirt: Eyes-Free Interaction while Running

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Figure 1: GestureShirt: (A) the shirt with various Velcro points to attach the sensors, (B) Running with the GestureShirt, making a "Swipe" gesture with the left hand to play the next track (C) Making a "Hold" gesture with the right hand to play/stop music

ABSTRACT

Running is a widely practiced sport, and technology is a prominent aspect of the experience. Runners often utilize music, tracking applications, and wearable sensors like heart rate monitors. Engaging with these systems while moving presents specific challenges, as the running motion can impede interaction capabilities. Standard devices such as smartphones, smartwatches, and fitness trackers are commonly used, yet their small buttons and touch screens necessitate slowing down for effective interaction. In this work we demonstrate GestureShirt, a shirt that allows for eyes-free interaction while running. The prototype shirt utilizes detachable distance sensors to capture simple gestures. In the demo, visitors can utilize the body's surrounding space to interact with music.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile devices; Gestural input.**

KEYWORDS

Gestures, Touchless Interfaces, Human-Computer interaction, Running

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1 INTRODUCTION

Running is one of the most prevalent forms of physical activity globally [8]. The incorporation of technology during the run has become standard and aims to enhance the running experience, serving to motivate, entertain, track, coach, and guide runners. Common commercially available technologies used during running include smartphones and a multitude of applications available on platforms like Google Play and the Apple App Store. Smartwatches, such as the Apple Watch, are accompanied with its specialized running applications. Despite their portability, these technologies are not optimized for interaction in motion due to their small buttons and touch screens, often resulting in the necessity to "stop-to-interact" [13]. Research indicates that even while walking, users are compelled to slow down by 25% when interacting with these devices [1]. Furthermore, Schildbach and Rukzio demonstrated that interaction efficacy is compromised even during walking [17]. The divided attention during movement poses potential risks to users [16], a challenge exacerbated during running due to higher speeds and more frequent environmental changes [16].

This work presents a gestural interface utilizing the space around the body for interaction. Leveraging our sense of proprioception [4], gestural interfaces offer the potential for eyes-free operation. Previous research has examined gestures during treadmill running, relying on the stable infrastructure provided by the treadmill [12]. In our work, we developed a wearable mobile technology prototype shirt equipped with standard ultrasound distance sensors to facilitate three distinct gestures during running, specifically for music control.

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2 RELATED WORK

Human-Computer Interaction (HCI) has extensively studied sports [5] and running over the past decade, as evidenced by numerous contributions to the field [9, 10, 14]. Technological support for running techniques, such as foot-striking [7] and breathing methods [22, 23], have been explored. Marshall advocates for mobile interactions that avoid the need to stop and interact with devices, a common issue with traditional smart devices [13]. Seuter supports this view by showing that interaction with smart devices affects running motion [19]. Consequently, researchers have investigated alternative interaction methods with technology during running, such as using feet [20] or toes [15].

2.1 Eyes-Free Interactions

Various studies have explored eyes-free interactions for input during movement, primarily using walking as the typical scenario. Wang et al. proposed eyes-free interaction by drawing on the hand palm [25]. Also movements with thumb were explored and captured by a ring [11]. Yeo and colleagues utilized the whole body as a surface for gestures [27]. Brewster et. al investigated head gestures, using nodding as an eyes-free input method [2]. Most studies do not report on the impact of these interactions on walking movement, although Brewster’s study noted a 30% reduction in normal walking speed [2]. Head-mounted displays have been used to capture hand tapping on the body [3]. Tapping the body has also been examined in the context of running. Here, studies evaluated the precision and comfort of various body locations for tapping during runs. Vechev et al. [24] found that the head and hands were the easiest to target and had minimal impact on running movement, while the upper chest and hips were less favorable. Hamdan and colleagues [6] focused more on the arms and abdomen, finding the abdominal area and wrists to be the most comfortable for tapping, though participants were less confident targeting the abdominal area compared to the shoulders. Overall, participants achieved high accuracy tapping specific locations on the arms (around 95%), but significantly lower accuracy on the abdominal area (around 35%).

In our work, we developed a prototype to investigate the space around the body for gesture-based interaction. Our system goes beyond touch-based interactions, enabling a broader range of interaction types.

2.2 Gestures while Running

Gestures offer significant potential for interaction during running. For example, Seuter et al. [18] allowed participants to explore gesture-based interactions with drones during runs, though these gestures were not implemented. Marin et al. [12] incorporated gestures into exergaming on a treadmill. In this study gestures were used as game elements, e.g. to collect coins. Vallat and colleagues [21] developed gestural controls for a treadmill using a Leap Motion sensor to capture hand gestures, which allowed runners to interact with the treadmill. They compared this with tactile interactions and gestures using the Myo bracelet, finding that runners preferred the wireless Myo gestures over the Leap Motion’s fixed interaction area, despite the Myo being limited to swipe interactions and unable to capture upward arm movements. Contrary to these approaches, we aim to facilitate gestural interaction without

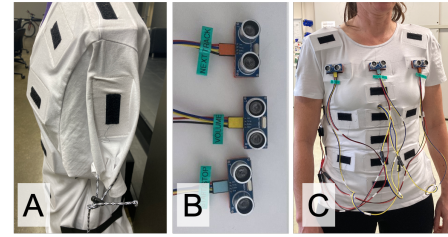


Figure 2: (A): Close-up of the sensor shirt, (B): Ultrasound distance sensors (HC-SR04) utilized for this demo, (C) Sensors attached to the shirt for initial testing

the need for fixed infrastructure like the Leap Motion, by developing a robust mobile wearable system. Additionally, we explore absolute mapped interactions, where the hand’s absolute position is mapped to functionality, differing from mid-air gestures that can be performed without regard to exact location. Our system supports various gesture types, including swipe gestures and distance-based hand-to-sensor interactions.

3 GESTURESHIRT PROTOTYPE

In this section we describe the design decision and technological parts of the GestureShirt. The goal was to develop a prototype that allows a fast attachment and detachment of various sensors on a running shirt. Additionally, the use case of operating a music player was designed with gestures for distance sensors.

3.1 Physical Prototype

Sensor positions should be in the front of the torso as well as on the shoulders as suggested by [24]). These areas are appropriate to be easily reachable for runners with their hand during running. We decided to have 18 positions on which the sensors shall be mountable (see Figure 3). Two on each shoulder, eight in the middle of the torso from the upper breast to the lower belly, and four on the left and four on the right side of the front torso. The the arm regions were avoided for sensor attachment due to their continuous movement during running. Distances in between the sensors are between 5 cm to 7 cm for the medium size t-shirt. For the other sizes the distances were adapted according to the size of the t-shirt. The back side of the shirt includes a pocket to hold the microprocessor for the sensors.

We decided to sew Velcro strips at the 18 different positions for the sensors (see Figure 4). This allowed us to easily affix sensor units to the shirt to provide flexibility in sensor placement. According Velcro strips were glued to the back side of the sensors. We attached cardboard squares on the back side of each Velcro strip (on the inside of the shirt) to increase stability of the sensor placement, making sure that the sensing direction deviated minimally. In order to allow a good fit for different sized runners, three prototype shirts were created in the sizes small, medium and large. The sensor locations were standardized based on the shirt’s dimensions, following the procedure described by [6]. Adjustable elastic bands were sewn to the sides of the shirts to fit each visitor’s body shape.

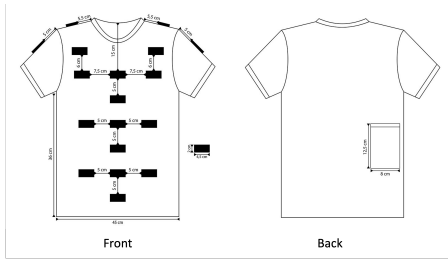


Figure 3: Front and back measurements of the GestureShirt in size M. Eighteen strips are spread over the shoulders and torso. The back side shows the integrated pocket to hold the microprocessor.

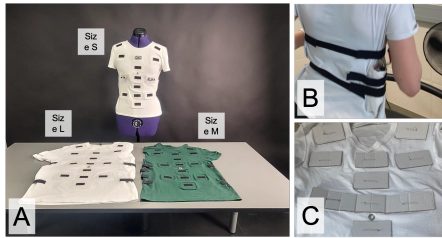


Figure 4: (A): Three t-shirt sizes to accommodate variety of people. (B): Elastic straps on the backside of the shirt to customize the tightness of the shirt, (C) cardboard squares were sewn to the inside of the shirt to increase stability of the sensors

3.2 Sensor Integration

As sensors we used HC-SR04 ultrasonic sensors. They were utilized for distance detection (see Figure 2). In order to allow to interact at three different positions with three different gestures we allowed up to three of these sensors simultaneously be attached to the shirt. The sensors were then wired to an ESP32 microcontroller. This was used to process sensor data, establish a Bluetooth connection, and send sensor readings to a smartphone. The "SR04" Arduino library was employed for sensor data acquisition. A reading interval of 300 milliseconds per cycle was chosen to balance gesture recognition accuracy and error minimization. A straightforward application was developed using MIT App Inventor¹, enabling local music control on the phone through the ESP32 microcontroller. The HC-SR04 ultrasonic sensors were operating on 5V DC voltage with an operating current of 15 mA. The manufacturing specs reported a measurement angle of 15 degrees and a sensing range between 2 cm and 4 m. When testing the sensors we were able to detect a hand in a range between 30 degrees horizontally (i.e. left and right) and 10 degrees vertically (i.e. up and down). Figure 5 visualizes this range.

3.3 Interaction Design

In order to have a use case for the GestureShirt we decided to design gestures for music controlling task during running. We

¹<https://appinventor.mit.edu/>

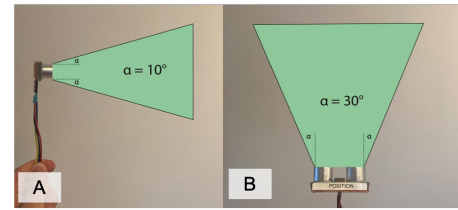


Figure 5: Angles of hand detection by the ultrasound distance sensors performed by researchers (A): the angle of detection vertically: 10 degrees (B): the angle of detection horizontally: 30 degrees

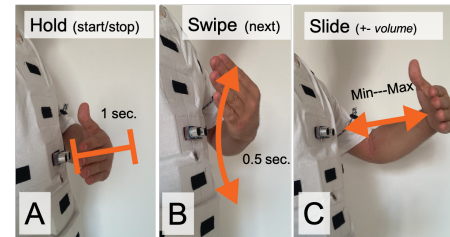


Figure 6: Three implemented gestures for the demonstration: (A) Hold: the user holds the hand in front of the sensor to start or stop the music. (B) Swipe: the runner swipes the hand over the sensor to play the next song. (C) Slide: the user moves the hand towards or away from the sensor to adjust the volume.

determined the most important functions during the run to be (a) play and stop music, (b) change to the next track in the play list, (c) change the volume. For each of these functionalities we designed and implemented gestures taking the Interaction Frogger framework as guiding principles [26].

The prototype integrates three gestural inputs. Figure 6 shows each of the gestures:

Hold: The runner holds the hand for 1 s in front of the sensor (within a range of 1 to 35 cm) to play or pause the music.

Swipe: The runner performs a short hand swipe (0.5 s) over the sensor range at a distance between 1 to 20 cm.

Slide: The runner moves the hand back and forward between 5 and 35 cm. The volume adjusts at an absolute scale; the closer to the body the lower volume, the further away from the body the higher the volume.

To ensure system robustness, each gesture was mapped to a distinct sensor, rather than multiple gestures being recognized by a single sensor.

4 DEMONSTRATOR POTENTIALS

The GestureShirt was designed to allow for a multitude of different studies that include interactions while being on-the-move. Here we want to discuss some potentials of the demonstrator.

Sensor Positioning. First, the shirt is equipped with Velcro fastener at various positions all across the frontal torso. It is very easy

and fast to position different types of sensors at different positions across the shirt. This allows for identifying the most convenient position for a user of a sensor. Potential studies range from doing AB testing with sensors at various positions to more exploratory studies, where user are allowed to position the sensors themselves and change their position easily. Additionally, the demonstrator allows for studies with a focus on different human factors, such as the influence of the size of an individual (Rem. we created three sizes of shirts: small, medium and large), left- or right-hand individuals, or gender specific differences.

Sensor Type. Second, the demonstrator does not only allow a free positioning of the sensor, but also to change different types of sensors and relate them to the position on the torso. In the use case described above we used ultrasonic distance sensors to operate the music player. It would be possible to equip the shirt with any kind of sensors or actuators for both input and output modality. For example, it could be used to position touch or pressure sensors on the Velcro fasteners, vibration actuators, inflatables, or LED's for peripheral displays. Studies performed by Hamdan et. al [6] and Vechev et. al [24] that evaluated different locations for tapping the body for interaction while running could be easily realized and execute in the field. Thus, studies are possible that do not only allow to explore various sensor positions, but also different types of modalities or interaction designs at different positions.

Feasibility Studies. Third, another type of studies that are enabled with the demonstrator are feasibility studies for various sensors and actuators for truly mobile interactions [13]. For example, it could be explored whether the weight or size of a sensor is appropriate to be worn during running, and what influence the type of movement (e.g. running speed) has on the interaction. Apart from that, the influence of contextual parameters such as the light conditions, temperature, or humidity has on accuracy of sensors.

Application Areas. Finally, with the demonstrator various applications could be enabled and tested. First, in the context of running other applications than operating a music player might be researched, such as the operation of a smart watch (e.g., change between modes such as heart-rate, distance, speed, etc.) or switching between phases of an interval training. Second, other application areas than running could be explored. This includes activities such as accepting or declining calls on a mobile phone while walking or changing between slides during a presentation.

5 CONCLUSION

In this paper we have presented a demonstrator that allows people to utilize the space around the body to interact while running. The demonstrated system allowed runners to interact through touchless gestures while running to allow for eyes-free truly mobile interactions during running. For demonstration purposes three interactions have been designed and mapped to music. The principle of the concept can be broadened to different contexts, e.g. to interact with different functionalities such as picking up the phone while walking, or changing slides in a presentation.

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